

Social Presence in VR Empathy Game for Children: Empathic Interaction with the Virtual Characters

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ABSTRACT

This paper discusses children's empathic interactions with the VR characters. The study and findings in this paper were derived from a larger research study that explored the design and evaluation of empathic experiences for young children (6-9 years old) in VR environments. We found similarities between how children interacted with the VR characters and with real people (i.e., cognitive empathy and emotional contagion). This provides initial insight into children's experience of social presence with the VR characters. We suggest follow-up research on connection between empathy and social presence to explore the ways to create empathic VR experiences for children.

Keywords: Social presence, educational games, children, empathy, empathy games.

Index Terms: J.5 [Arts and Humanities]: Computer Applications; K.3.1 [Computer Uses in Education]: Computers and Education—Computing Milieux

1 INTRODUCTION

This research initially focused on exploring how to create empathic experiences for children. De Waal [6] discusses different ways how people tend to express empathy and respond to others' emotions: (1) cognitive empathy and (2) emotional contagion. First, people who express cognitive empathy tend to only partially or not at all match the emotional state of a person they are empathizing with. This suggests that people expressing cognitive empathy would not necessarily express similar emotions that would match another person's emotional state. In addition, De Waal [6] suggests that it is more likely that a person who experiences cognitive empathy and does not match the emotional state of another person has a predisposition for helping. Second, people who express emotional contagion tend to match somebody else's emotions [6]. De Waal [6] states that when a person matches the emotional state of another person, that gives them personal distress and emotional transfer.

While exploring the ways to create empathic experiences for children (6-9 years old) in VR, we found evidence of social presence in how children interacted with the VR characters. Social presence is defined as "a phenomenon when a person senses a connection with another being and believes that they interact with another human being" [9]. The research found that the mutual gaze, distance, and behavior of a virtual human affects the level of social presence that a person can experience while interacting with a virtual human [1][5][3]. Daher [5] investigated whether a high social presence can be built between a person and a virtual human. He reported that short social priming, such as using a pre-

planned conversation between the facilitator and a virtual human before a VR experience with a virtual human, might significantly increase a person's attention, mood, co-presence, and social presence towards a virtual human.

There are still several open research questions on social presence in VR. For example, there are questions related to understanding how the level of realism, appearance, and behavior of virtual humans affect people while they interact with virtual humans in a VR environment. Also, there is still uncertainty about the nature of a social presence in VR. Bailenson [2] ponders whether the nature of social presence is purely affective or has cognitive aspects. In this paper, we present an insight into the nature of social presence in a virtual reality empathy game.

We found similarities between how children interacted with the VR characters and with real people: (1) cognitive empathy and (2) emotional contagion. These findings provide initial insight into affective and cognitive nature of social presence. In this paper, we describe and discuss these patterns.

2 STUDY AND METHODS

The study and findings in this paper were derived from a larger research study that explored the design and evaluation of empathic experiences for young children in VR environments. We designed a VR Empathy Game: "Why Did Baba Yaga Take My Brother?" to engage children in an immersive and open-ended game environment that provided them with opportunities to interact with VR characters representing different emotions [8]. For example, the two antagonists in the game, Goose - represented the emotion of fear and Baba Yaga represented the emotion of anger (fig. 1).

Children were able to interact with the VR characters using the following features: (1) dialog, (2) memories, and (3) glasses. Dialog feature provided an opportunity to communicate with the game characters using dialog orbs. Children could view memories of each character to understand how the character's past experiences shaped their personalities and emotional behavior. By wearing each character's perspective glasses, the player can see the world from the character's perspective. Wearing glasses alters the world's colors, sounds, and other characters.



Figure 1: Baba Yaga Character.

To investigate how children understood and interacted with the VR Empathy Game characters and environments, we designed a

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90-minute study. Each session included three 30-minute phases: (1) pre-surveys; (2) VR Empathy Game experience; and (3) post-surveys and post-interview. We conducted this study individually with 14 participants (6 girls and 8 boys of 6-9 years old) and collected various data including interviews, observations notes, empathy scales surveys [10]. Our study was approved by IRB #201600947, which required us to collect signed parent consent forms and oral children's consent. We used a case study methodological approach and analyzed the data using inductive coding and thematic data analysis [7].

3 FINDINGS

While exploring the ways to create empathic experiences for children in VR, we found similarities between how children interacted with the VR characters and with real people. Based on observations and interview answers, we identified two different behavioral patterns: pattern#1: cognitive empathy (6 children) and pattern#2: emotional contagion (3 children). In this section, we will describe and discuss these initial patterns using case Tima for representing pattern #1 and case Camila for representing pattern #2.

3.1 Pattern One: Cognitive Empathy

Tima's case provided evidence of her engagement in the interactions with the game characters: (1) she was interested in interacting with the characters and helping them, (2) she expressed empathy. From the other side, we observed Tima being emotionally distant from the game characters and staying calm while playing. As Tima stated in the postgame interview, the VR experience was for her "like real but not real." This phrase of hers can describe Tima's style of interacting with the game characters. Tima was engaged in interacting with the game characters. At the same time, Tima kept her emotional distance from the characters and was not affected by their emotions.

We found the explanation of Tima's behavior in the literature. While discussing cognitive, De Waal [6] states that people who express cognitive empathy tend to only partially or not at all match the emotional state of a person they are empathizing with. De Waal [6] explains that emotional state matching "can be arrived at in a top-down fashion, involving emotional circuits to a lesser extent." This suggests that people expressing cognitive empathy would not necessarily express similar emotions that would match another person's emotional state. Our observation of Tima's interaction with the game characters supports de Waal's [6] findings. In addition, De Waal [6] suggests that it is more likely that a person who experiences cognitive empathy and does not match the emotional state of another person has a predisposition for helping. We observed it in Tima's behavior. For example, while observing one of the characters' cryings, Tima approached them and hugged them before asking them any questions.

3.2 Pattern Two: Emotional Contagion

Camila experienced distress from matching emotions with Goose (i.e., fear), which stopped her from playing the game. In the psychology literature: such phenomenon of matching somebody else's emotions is called emotional contagion [6]. As well as the Goose character, Camila also experienced fear. De Waal [6] states that when a person matches the emotional state of another person, that gives them personal distress and emotional transfer. Both of them were observed in Camila. First, we observed how Camila was emotionally distressed as her behavior changed while wearing the glasses. We suggest that the gameplay experience affected Camila's emotional state, particularly interaction with Goose and his Glasses. While wearing the Goose's glasses, Camila's actions

got more quick and abrupt, and she seemed not to feel comfortable while wearing his glasses. She was noticeably more quickly than usual, turning around, trying to see what was happening around her. Second, from Camila's words, we found that Camila got scared, which matched the state of the Goose character. De Waal [6] called such phenomena of emotions matching with the term of emotional transfer.

4 DISCUSSION AND CONCLUSION

This paper provides insight on (1) children's interactions with VR Empathy Game characters and (2) the nature of social presence in VR environments. Our findings provide insight into types of child-character empathic interactions (i.e., expressing emotional contagion or cognitive empathy towards the game characters). For example, Camila experienced distress from emotional contagion with Goose, and she could not finish the game. This suggests that interactions between children and VR Empathy Game characters can replicate interactions between humans, particularly how emotional contagion (sharing others' distress) affects their ability to express prosocial behavior. These results expand an understanding of the nature of children's communication with the VR game characters. VR researcher Bailenson [2] stated that there is still uncertainty about the nature of a social presence in VR and pondered whether the nature of social presence is purely affective or has cognitive aspects. We plan to conduct follow-up research on how de Waal's [6] cognitive empathy model applies to the VR game characters' social presence. Such exploration of the application of de Waal's (2000) concept of interaction between humans onto interaction between a human and a VR game character may provide more understanding of the nature of social presence.

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